

ELLIAS SITHOLE

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PROFESSIONAL SUMMARY

Data and analytics professional with an MSc in Pure Mathematics and experience across data analysis, municipal revenue analytics, machine learning, and business operations. Strong in Python, SQL, Power BI, Excel, and Power Query, with hands on experience building dashboards, data pipelines, and operational reports. Proven ability to support telecom, municipal, supply chain, finance, and operational teams with data driven insights and automation.

PROJECTS

- Telco Churn Prediction Pipeline** , Python | FastAPI | Streamlit | Docker | docker-compose
- Built an end to end ML pipeline using Python and TensorFlow to predict customer churn on structured telecom data.
 - Implemented reusable preprocessing, model training, and inference layers
 - Deployed APIs and dashboards using FastAPI and Streamlit with Docker
- Inventory Demand Forecasting System** , Python | scikit-learn | FastAPI | Streamlit | MLFlow | Github Actions
- Built an end-to-end demand forecasting pipeline (ingestion → transformation → training → evaluation) with artifact tracking
 - Deployed prediction + explainability endpoints (FastAPI) and an interactive Streamlit UI
 - Added SHAP explainability, automated tests, CI workflow, and Dockerized services for reproducible deployment

KEY SKILLS

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|---|--|
| Programming and Querying <ul style="list-style-type: none">Python, SQL, JavaScript | BI and Analytics <ul style="list-style-type: none">Power BI, Power Apps, Excel, SSMS, Power Query |
| Data and Machine Learning <ul style="list-style-type: none">Pandas, NumPy, scikit learn, TensorFlowMachine learning, regression, classification, model evaluationDeep learning fundamentals, NLP fundamentals, large language models, prompt engineering | Engineering and Tools <ul style="list-style-type: none">Git, GitHub, Docker, FastAPI, Streamlit, GitHub Actions |

EMPLOYMENT HISTORY

- Senior Data Analyst**, Ntiyiso Consulting Group 02/2026 – Present
Midrand
- Analyze billing, meter reading, payment, and field work data across City of Johannesburg, Lesedi Local Municipality, Emfuleni Local Municipality, and Mogale City Municipality.
 - Build Excel and Power BI dashboards to track collections, field actions, debtor classifications, payment trends, and meter reading performance.
 - Use SQL, Excel, Power Query, and Power BI to identify anomalies, unpaid accounts, reading exceptions, and revenue recovery opportunities.

- Prepare stakeholder reports and management summaries highlighting key findings, risks, and recommended follow up actions.

Meter Reader Analyst, MMR JV Pathfinders
08/2025 – 01/2026
Vanderbijlpark

- Analyzed electricity consumption data using SQL and Excel to detect anomalies, leaks, and irregular usage patterns.
- Built Excel and Power BI dashboards for management reporting.
- Improved data accuracy through validation and reconciliation checks.

Intern Junior Developer, NVisionIT
11/2024 – 01/2025
Hybrid

- Assisted in developing commercial web applications.
- Collaborated with development teams on various projects.
- Worked with structured data inputs and APIs in commercial web applications.
- Collaborated using GitHub and Jira in an agile environment.

PepsiCo

Network Transformation (Network Modeler)
01/2023 – 12/2023
Isando, Kempton Park

- Built and validated network models using SQL and Power BI to support operational decisions.
- Supported cross functional teams with data driven insights for supply chain and finance across multiple business units.
- Validated input data for accuracy.

Finance & Revenue Management Graduate
08/2021 – 01/2023
Johannesburg, Rosebank

- Completed 18-month graduate program with rotations in R&D, Supply Chain, Commercial, and Finance.
- Worked on projects like Sustainability WECA and Brand Ladder Pricing Index.
- Contributed to ESG and pricing projects using structured datasets.

ACADEMIC BACKGROUND

MSc in Pure Mathematics, University of the Witwatersrand
02/2020 – 12/2021
Johannesburg

- Research dissertation in Graph Theory titled '**A Derived Graph: Triangle Graph of a Graph**', supervised by Prof Eunice Mphako-Banda.
- Strong focus on graph structures, abstraction, and mathematical reasoning applicable to algorithms and data modeling

BSc (Hons) in Pure Mathematics, University of the Witwatersrand
02/2019 – 12/2019
Johannesburg

- Focused on Advanced Number Theory.
- Research project: Generating Functions in Number Theory.

BSc in Computational and Applied Mathematics and Mathematics, University of the Witwatersrand
02/2015 – 12/2018
Johannesburg

- Majors: Pure Mathematics, Applied Mathematics, Physics, Chemistry.

ACHIEVEMENTS

Contributed to a cross-functional project that improved company ESG rating.

Improved delivery productivity by 8 percent through data driven schedule redesign.

Contributed to over R100 million in COJ municipal debt recovery within two months.

CERTIFICATIONS

**Microsoft Certified: Power
Platform Functional Consultant
Associate**
2024

**Microsoft Certified: Power
Platform Developer Associate**
2024

**Microsoft Certified: Dynamics
365 Sales Functional
Consultant Associate**
2024